

CyberDefense Pro 2.0

2.6.2 Exploring the Kill Chain

This lesson covers the following topics:

- Kill chain model
- Seven phases of an attack

Kill Chain Model

There are several models for describing the general process of an attack on system security. These steps are often referred to as a kill chain, following the influential white paper, *Intelligence-Driven Computer Network Defense*, commissioned by Lockheed Martin.

Kill chain analysis can identify a defensive course of action to counter the progression of an attack. Security teams must understand the kill chain stages in order to recognize how far suspicious activity has progressed. Additionally, the kill chain shapes analysis techniques as different approaches are suited to different stages. As an analyst, it is vital to identify suspicious activity as early in the kill chain as possible.

Seven Phases of an Attack



▼ Description

The image is a flowchart depicting the typical stages of a cyber attack, known as the "kill chain." There are seven circular icons, each representing a specific phase. Arrows indicate the sequence from one phase to the next:

1. **Reconnaissance** – represented by a magnifying glass; this step involves attackers gathering information about the target.
2. **Weaponization** – shown as a bomb; here, attackers create a deliverable payload (like malware).
3. **Delivery** – depicted by an envelope; the payload is sent to the target via email, web link, or another delivery method.
4. **Exploitation** – shown with an explosion symbol; the weapon is triggered, exploiting a vulnerability in the target system.
5. **Installation** – illustrated with a software download icon; malware is installed on the victim's system.
6. **Command & Control (C2)** – represented by a networked computer icon; the compromised system connects to an attacker's remote server for further instructions.
7. **Actions on Objectives** – depicted by a hand clicking a target; attackers achieve their final goals, such as data theft or system disruption.

The path connecting the icons is staggered, visually reinforcing the step-by-step process of an attack.

The kill chain includes the seven phases of an attack as described in the following table:

Stage	Description
Reconnaissance	The adversary gathers information about the network using network probes, open-source intelligence (OSINT), and social engineering. The aim is to map the total attack surface and identify potential attack vectors. Targets are then researched, identified, and ultimately selected.
Weaponization	The attacker identifies a method by which identified vulnerabilities can be exploited, often through weaponized code, such as carefully crafted scripts, custom malware binaries, compromised websites, social engineering, and other methods. The exploit code is coupled with a payload that will assist the attacker in maintaining and extending covert access.
Delivery	The weaponized code is inserted into the environment using a selected attack vector, such as email attachment, phishing website/download, USB media, and so on.

Stage	Description
Exploitation	This results in the weaponized code running on the target system. For example, a phishing email may trick the user into running the code, while a drive-by download would execute on a vulnerable system without user intervention.
Installation	The payload is successfully installed on the target system using methods to remain undetected and achieve persistence.
Command and Control	During this stage, the payload establishes a connection to a remote server, enabling the attacker to connect to the target and download or fabricate additional information. (Also known as C&C or C2.)
Actions on Objectives	The adversary uses the compromised system to achieve or progress towards goals, such as data exfiltration, DoS vandalism, or escalating access across the target network or other connected networks.

Copyright © CompTIA, Inc. All rights reserved.